

Halvor Hafnor

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SUMMARY

Embedded software and machine learning engineer. I build classification pipelines that run on real sensor data and constrained hardware, from collection and feature extraction through to on-device inference. Two years at Aerendir Mobile on embedded Linux and Android, with a shipped automotive product and a pending patent, while finishing an M.S. in Computer Engineering at UC San Diego.

EXPERIENCE

Embedded ML Engineer

Aerendir Mobile

Jul. 2024 – Present

Mountain View, CA

- Developed embedded machine learning pipelines for classifying biometric motion signals from inertial sensors on **Linux**-based mobile devices.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the motion dataset behind the company's biometric authentication work: **400 participants**, each recorded holding the phone in three postures, sitting, standing and walking, from the accelerometer and gyroscope.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel**. Evaluated ST's on-sensor **Machine Learning Core**, then replaced it with a threshold classifier over hand-engineered vibration features running in the IMU's own firmware. Subject of a pending patent naming me as an inventor.

Electrical Engineering Intern

ABB Electrification

Jun. 2021 – Sep. 2023

Skien, Norway

- Assembled and wired industrial electrical cabinets and control panels based on engineering schematics.
- Supported production and testing of custom electrical systems for industrial clients.

EDUCATION

University of California - San Diego - San Diego, CA

Computer Engineering, M.S.

Jan. 2026 – June 2027

Computer Engineering, B.S. Major GPA: 3.9

Sep. 2024 – June 2026

Honors

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

De Anza College - Cupertino, CA

Computer Science & Engineering

GPA: 4.0

Academic and Technical Roles

Sep. 2022 – June 2024

- Math & Science Tutor and Computer Science Teaching Assistant for calculus, linear algebra, discrete math, physics, and C++, Java and Assembly.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly

Embedded: Embedded Linux, Android, QML/Qt, inertial sensing, on-sensor inference, ST IMU firmware, Docker

Signals & ML: Feature extraction and classification from motion data, OpenCV, NumPy, SciPy, YOLOv8

Systems: Node, SQLite, WebSocket protocols

Agentic Development: Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

PROJECTS

TabTerm, a macOS terminal in Chrome tabs

- A **Chrome extension** over a background **terminal daemon** that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search and project-scoped saved commands.

Laser targeting on an autonomous car, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5. **YOLOv8** for detection and an **OAK-D** stereo camera for depth, converted through the camera's field of view into pan and tilt angles so two servos keep a laser on the nearest person as they move. Depth is sampled as the median of a patch, since a single bad pixel throws the aim. Deployed in **Docker** on the vehicle.